

A Dual-Agent Framework for Translating Unstructured Biological Experimental Protocols with Cross-Verification in Self-Driving Laboratories

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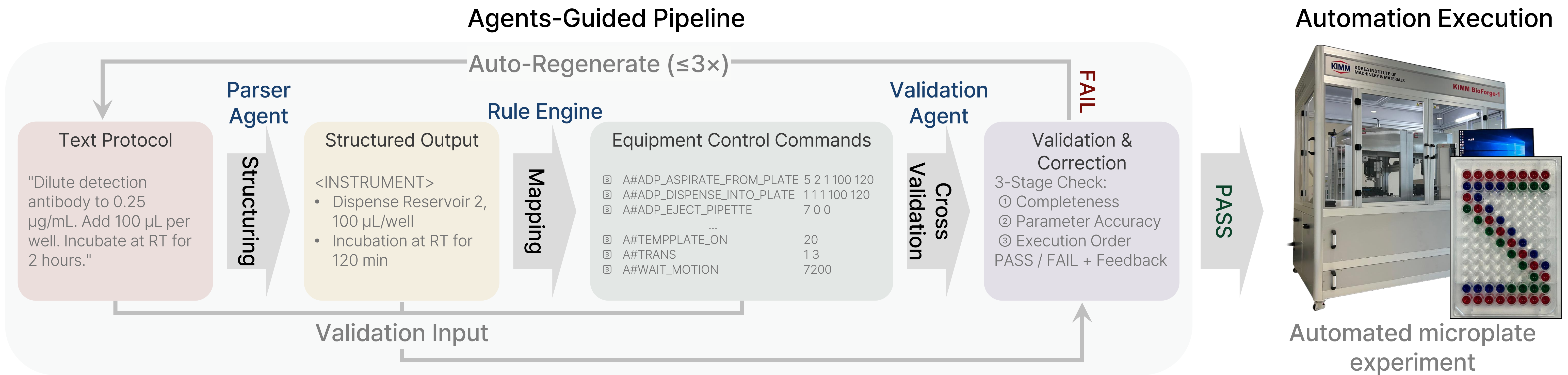
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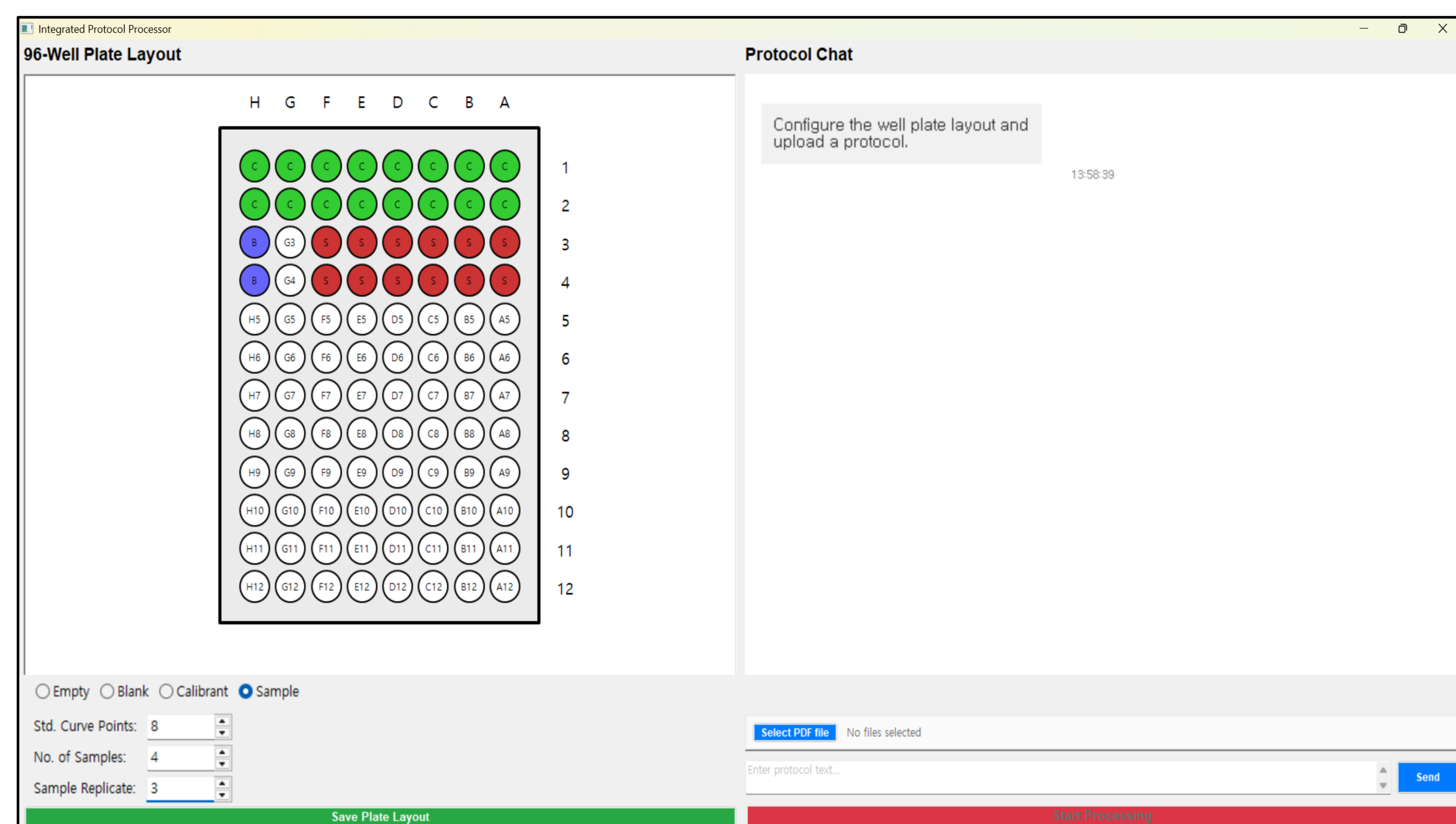
Introduction

1. Natural-language biological experimental protocols are inherently unstructured, containing implicit procedural knowledge and ambiguous instructions that hinder direct automation.
2. Converting such protocols into pre-defined equipment commands remains a fundamental challenge for realizing self-driving laboratory systems.
3. A dual-agent framework is proposed, in which a Parser Agent structures the protocol and a Validation Agent performs iterative cross-verification with auto-regeneration.
4. The framework was deployed on a microplate-based experiment automation platform and validated, including total protein quantification of various Fetal Bovine Serum(FBS) batches.

Dual-Agent Pipeline for Automated Translation and Cross-Verification of Natural-Language Protocols



Proposed System: UI Interface and Hardware Platform



- User-defined well allocation for Std. Curve Points, Samples, and Replicates
- Chatbot-style protocol input supporting PDF upload or direct text entry
- The input protocol and well allocation are passed directly to the dual-agent framework

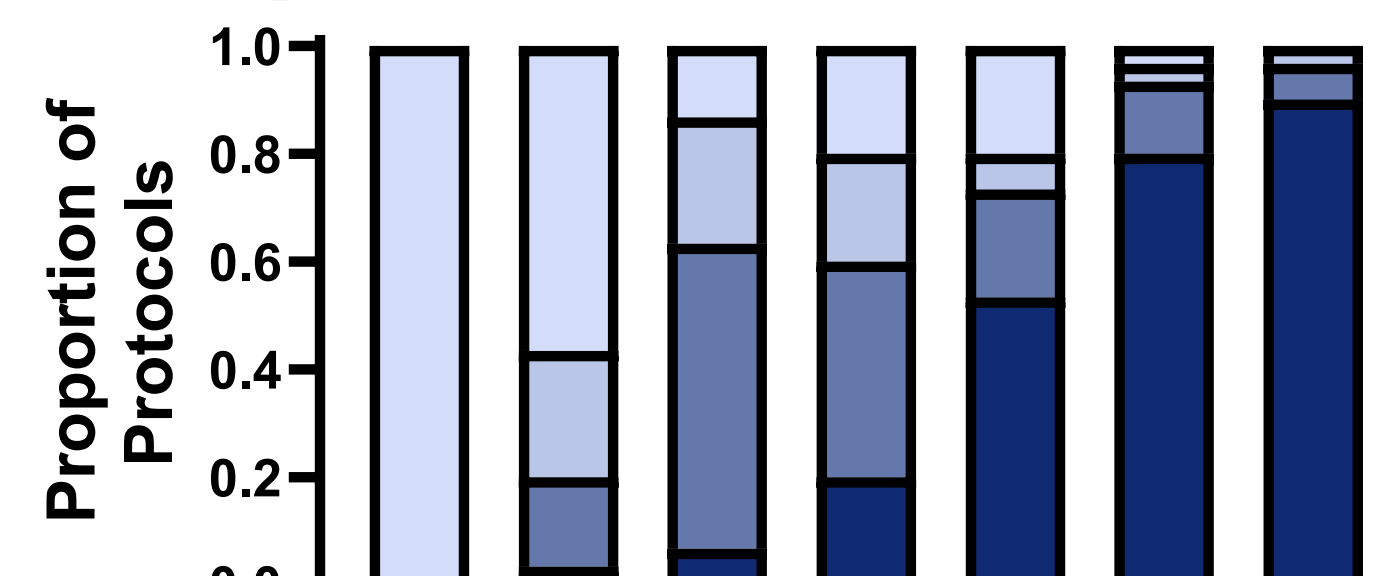


- Bio-experiment automation platform for direct execution of command sequences

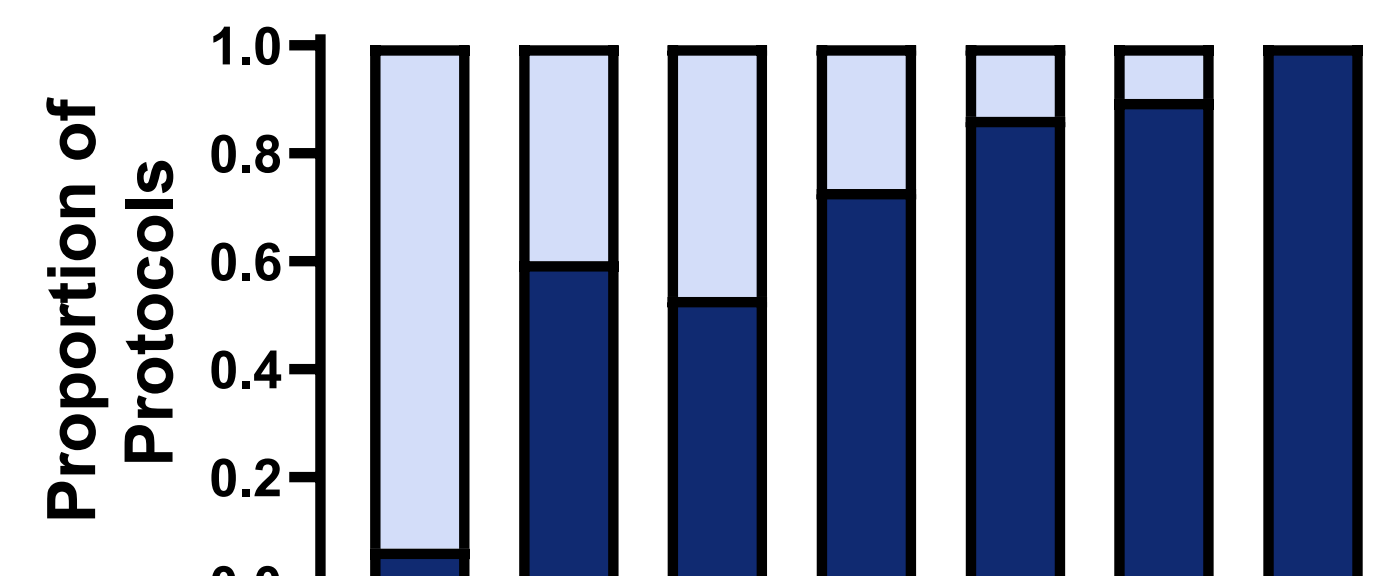
Dual-Agent Evaluation Results

- 7 Parser Agents × 3 Validation Agents
- Claude Sonnet 4.6: precise, actionable feedback → consistent accuracy improvement
- Llama-4-Maverick & GPT-5: vague, insufficient feedback → accuracy plateaus or degrades

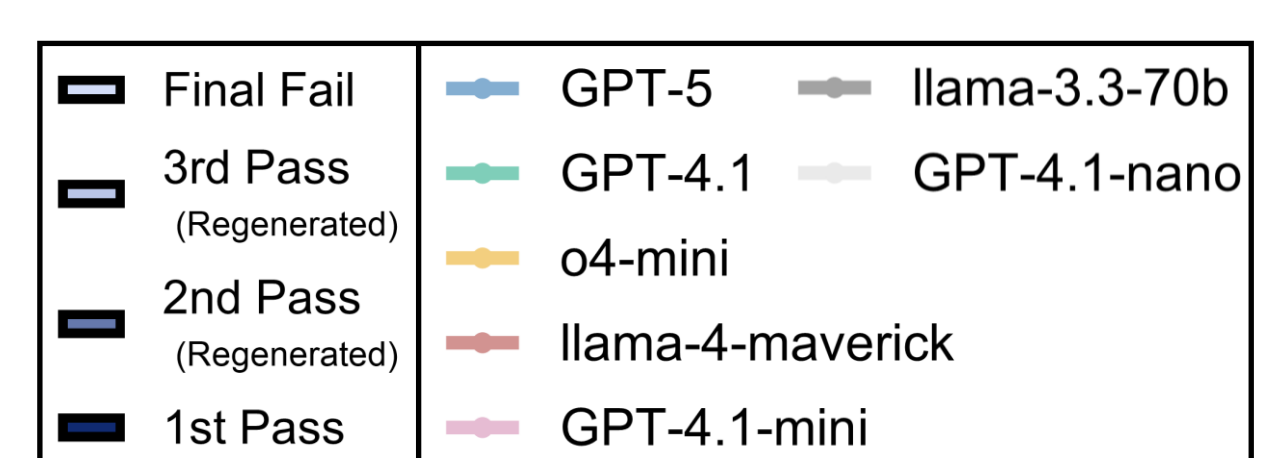
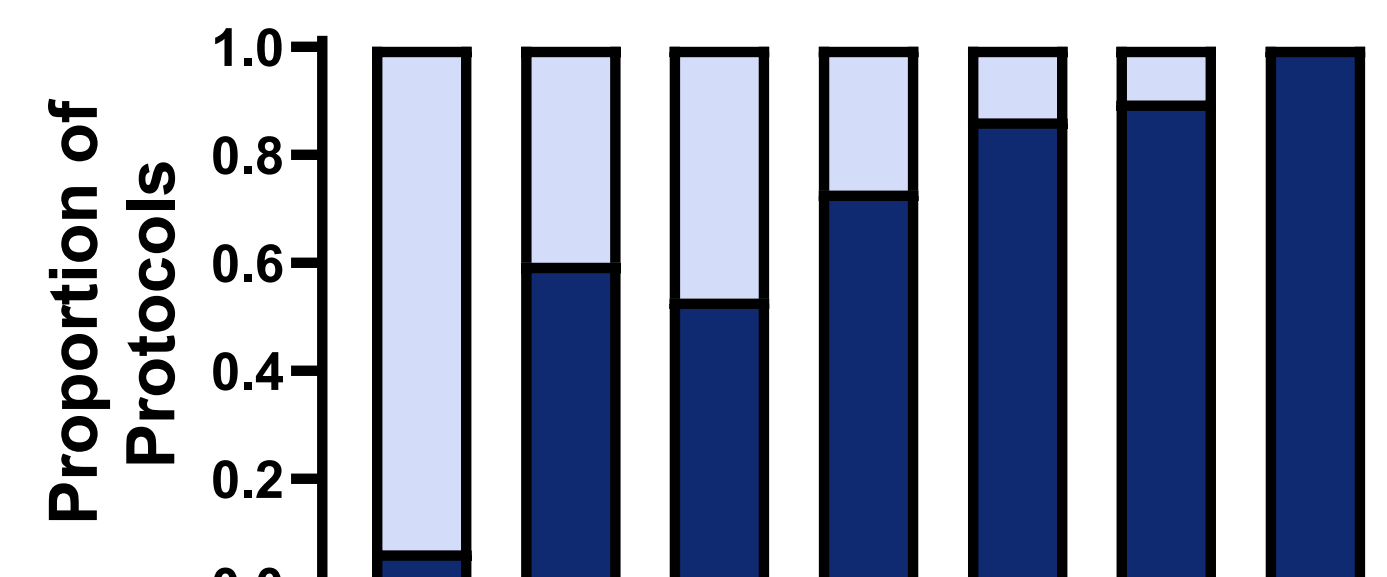
Protocol Pass Rate by Parser Agent [Validation Model: **Claude Sonnet 4.6**]



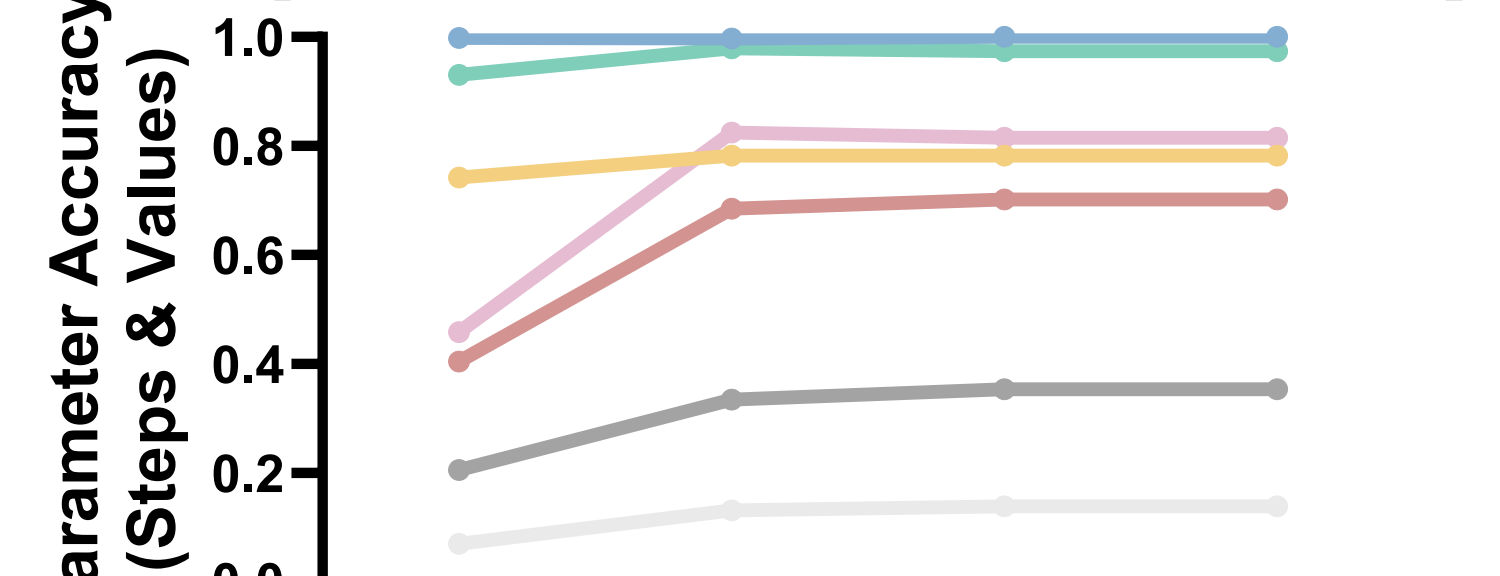
Protocol Pass Rate by Parser Agent [Validation Model: **llama-4-maverick**]



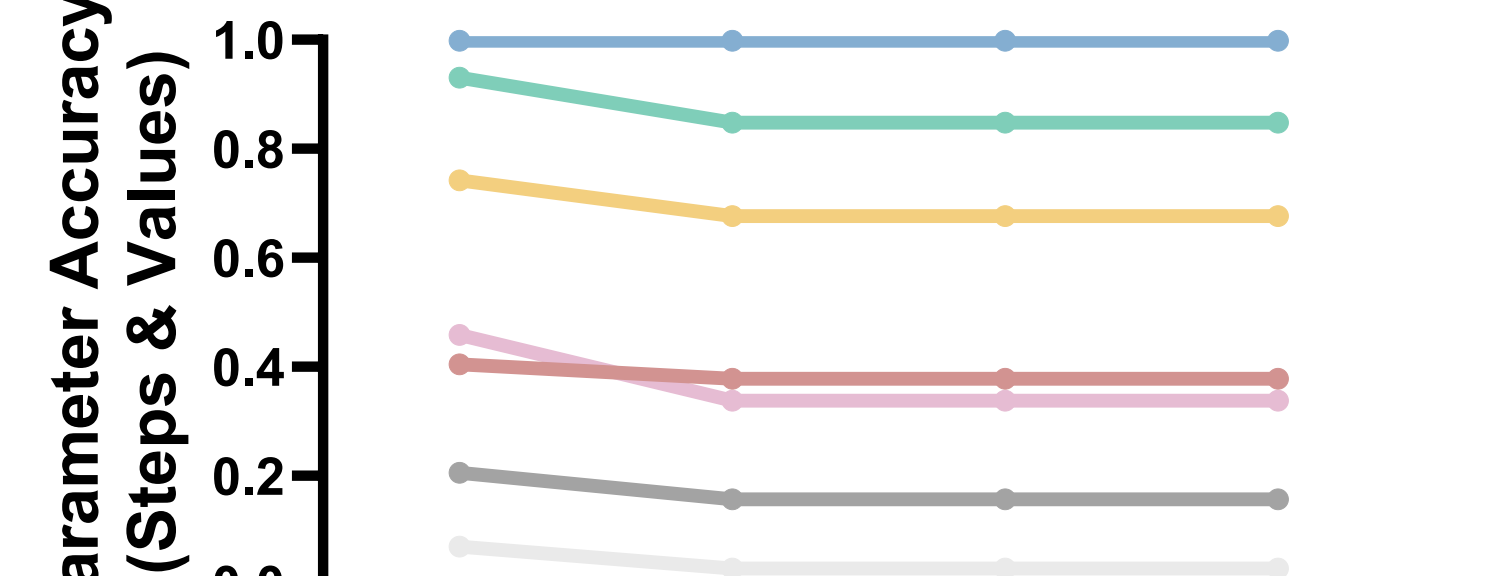
Protocol Pass Rate by Parser Agent [Validation Model: **GPT-5**]



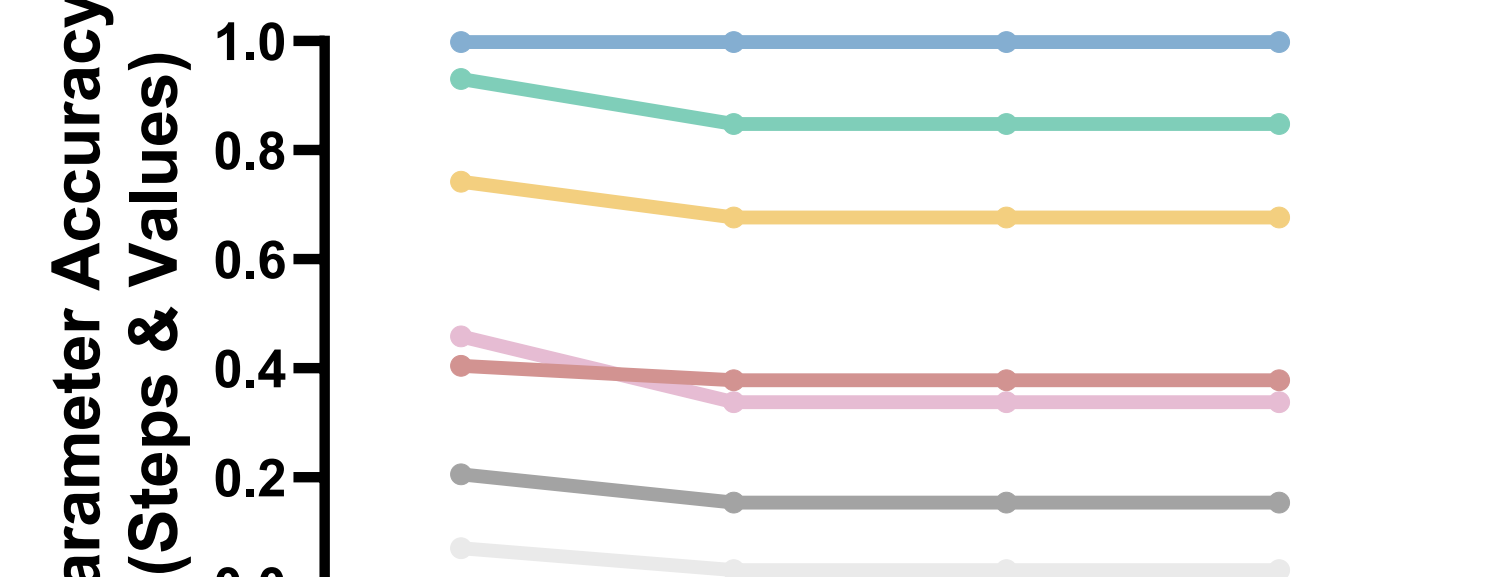
Protocol Accuracy per Validation [Validation Model: **Claude Sonnet 4.6**]



Protocol Accuracy per Validation [Validation Model: **llama-4-maverick**]



Protocol Accuracy per Validation [Validation Model: **GPT-5**]



Bio-Experiment Automation

Experiment ①	Experiment ②
Unstructured Protocol "Add 100 µL of Blank, Calibrant, or Sample to the appropriate wells. Incubate for 5 minutes at room temperature."	Unstructured Protocol "Pipette 10 µL of Blank (distilled water), Calibrant (5–25 µg/mL BSA), or serum Sample into target wells. Add 200 µL of Bradford Reagent per well. Shake at 300 rpm for 30 seconds. Incubate for 10 minutes at room temperature. Read absorbance at 595 nm and calculate total protein concentration against the standard curve."
Execution Result 	Execution Result

※ Parser Agent: GPT-5 | Validation Agent: Claude Sonnet 4.6 | 1st Pass

Conclusion

1. Unstructured natural-language protocols were accurately translated into pre-defined equipment commands, achieving high fidelity in both step ordering and parameter values.
2. An auto-regeneration framework with iterative cross-verification (up to 3 rounds) was proposed, improving translation quality and enabling applicability to local LLMs.
3. Validation Agent quality proved critical: Claude Sonnet 4.6 delivered structured feedback that drove consistent accuracy gains across iterations.
4. Real bioassay experiments, including serum protein quantification, validated the framework's practical utility for laboratory automation.